CS361 — Cybersecurity Project

Checkpoint 3

Defensive Controls & Validation Progress

AI-Assisted Phishing Detection for Healthcare Email Systems

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GitHub: https://github.com/SergioOlivas1/CS361

1. Executive Summary

This Checkpoint 3 report documents the defensive controls implemented, simulated, and evaluated for our AI-assisted phishing detection system in a healthcare clinic email environment. Building directly on the Checkpoint 2 baseline, our team upgraded the detection system from a 5-test, keyword-only prototype (v1) to a 15-test, multi-control system (v2) with measurably improved accuracy and new defensive capabilities.

The v2 system introduces six new defensive controls: obfuscation-resistant keyword detection, display-name spoofing detection, automated quarantine flagging, CSV-based logging, expanded keyword and lure libraries, and a broader test suite with false-positive checks. Validation testing across 15 cases produced 93.3% accuracy, 100% precision, and 88.9% recall — compared to an estimated 60% accuracy on the original 5-case baseline.

Two edge cases (TC-08, TC-09) revealed residual scoring gaps that are documented in the updated risk register as areas for future improvement. All test output is captured in an automatically generated log file (phishing_log.csv), providing the traceability and auditability that were absent in the baseline.

2. Control Implementation Summary

The following table maps each new defensive control to the vulnerability it addresses, the implementation approach, and the threat category targeted.

ID	Control	Vulnerability Addressed	Implementation	Threat
C-01	**Obfuscation-Resistant Detection**	Phishing keyword evasion via character substitution (v3rify, acc0unt)	Added regex patterns matching leetspeak/mixed-character variants of 6 high-signal keywords	AI evasion
C-02	**Display-Name Spoofing Detection**	Attacker sets display name to "Clinic Billing" but sends from external domain	Added check: if display name contains clinic-related terms but sender domain is not in trusted whitelist, +25 pts	Social engineering
C-03	**Quarantine Flag**	No action taken on Phishing classification beyond printing output	Added quarantined=True field on DetectionResult for all Phishing classifications — foundation for system integration	All phishing threats
C-04	**CSV Logging**	No record of what was analyzed, classified, or flagged	Every analysis writes a timestamped row to phishing_log.csv with sender, subject, score, classification, quarantine status, and all flags	Traceability, audit
C-05	**Expanded Keyword Library**	v1 had 15 phishing keywords and 8 healthcare lures	v2 has 21 phishing keywords (+6) and 15 healthcare lures (+7), covering Medicare/Medicaid language, EOB, lab results, referrals	Detection coverage
C-06	**Extended URL Pattern Detection**	v1 checked 4 URL patterns	v2 checks 7 URL patterns (+3), adding login-page subdomain patterns, secure-update fakes, and verify-account URLs	Phishing link delivery

2.1 Why These Controls Were Selected

Controls were selected based on two criteria: (1) direct mapping to a risk identified in the Checkpoint 2 risk matrix, and (2) feasibility within the project scope and timeline.

- C-01 (Obfuscation) addresses the highest-rated risk in CP2 — AI evasion. Attackers commonly substitute characters to evade keyword filters. This control directly closes that gap with minimal implementation cost.
- C-02 (Display-Name Spoofing) addresses the social engineering threat. Display-name attacks are the most common phishing technique because they exploit how email clients render sender names, not addresses.
- C-03 and C-04 (Quarantine + Logging) address the baseline evidence gaps identified in CP2. Without quarantine flags and a log, the system has no audit trail and no mechanism to act on its own classifications.
- C-05 and C-06 (Expanded Libraries) directly improve recall — the proportion of actual threats the system catches — by covering more of the vocabulary attackers use in healthcare-specific campaigns.

3. Evidence of Defensive Work

3.1 Filtering Logic — Obfuscation Detection (C-01)

The following patterns were added to phishing_detector_v2.py to catch character-substitution evasion attempts. Each match adds +15 points to the risk score.

```
OBFUSCATION_PATTERNS = [
       (r"v[e3]r[i1]f[y!]",       "obfuscated 'verify'"),
       (r"acc[o0]unt",            "obfuscated 'account'"),
       (r"p[a@]ssw[o0]rd",        "obfuscated 'password'"),
       (r"cl[i1]ck\s+h[e3]r[e3]","obfuscated 'click here'"),
       (r"susp[e3]nd[e3]d",       "obfuscated 'suspended'"),
       (r"urg[e3]nt",             "obfuscated 'urgent'"),
]
```

Validation: TC-10 subject "V3RIFY YOUR ACC0UNT NOW" triggered 3 obfuscation flags and was correctly classified as Phishing (80/100). This email scored 0 in v1 because no exact keyword match was found.

3.2 Display-Name Spoofing Detection (C-02)

The following logic was added to the detection engine to catch emails where the display name implies a trusted sender but the actual email domain is external.

```
# Trusted sender domain whitelist
TRUSTED_DOMAINS = { "yourclinic.org", "clinic-group.org" }

# Display-name spoof check
if email.display_name:
       name_lower = email.display_name.lower()
       sender_domain = email.sender.split("@")[-1].lower()
       clinic_terms = ["clinic","hospital","health","medicare","billing"]
       if any(t in name_lower for t in clinic_terms):
       if sender_domain not in TRUSTED_DOMAINS:
              score += 25
              flags.append("Display-name spoofing detected")
```

Validation: TC-07 (display name "Clinic Billing Department", sender billing@healthcare-updates.net) and TC-14 (display name "HealthFirst Insurance Claims", sender claims@insurance-portal.ml) both triggered this control and were classified as Phishing.

3.3 Quarantine Flag Design (C-03)

All emails scoring 60+ now receive quarantined=True on their DetectionResult object. In a production deployment this flag would trigger routing to an isolated quarantine folder instead of the employee inbox. In the current prototype it is logged and printed with every output.

```
# In analyze_email():
if score >= 60:
       classification = "Phishing"
       action       = "Quarantine immediately. Do not deliver."
       quarantined  = True

# Output now shows:
#   QUARANTINED     : YES
```

3.4 Logging Design (C-04)

Every call to analyze_email() appends a row to phishing_log.csv. The schema captures all information needed for forensic review and audit compliance.

```
CSV Schema:
  timestamp         — datetime of analysis
  sender            — full sender address
  subject           — email subject line
  risk_score        — numeric score 0-100
  classification  — Safe / Suspicious / Phishing
  quarantined       — True / False
  flags             — pipe-delimited list of all triggered flags
```

Sample log output from the 15-case test run (first 3 rows shown):

```
timestamp,sender,subject,risk_score,classification,quarantined,flags
2026-04-19 ...,billing@yourclinic.org,Appointment confirmed...,0,Safe,False,
2026-04-19 ...,no-reply@medicare-update.xyz,URGENT: Patient
portal...,100,Phishing,True,
             Phishing keyword: confirm your identity | Healthcare lure: patient
portal |
             Suspicious sender | Suspicious URL | Urgency signal | Obfuscation
(x5)
2026-04-19 ...,support@insurance-verify.click,Insurance
Verification...,90,Phishing,True,
             Phishing keyword: update your billing | validate your credentials |
...
```

4. Validation Evidence

4.1 Full Before/After Test Table

v1 scores shown for TC-01 through TC-05 (the original baseline test cases). TC-06 through TC-15 are new cases added in v2. Pass/fail is measured against the expected classification.

ID	Sender / Subject	v1 Score	v1 Class	v2 Score	v2 Class	Expected	Pass	New Controls
TC-01	billing@yourclinic.org / Appointment confirmed	0	Safe	0	Safe	Safe	✓	—
TC-02	no-reply@medicare-update.xyz / URGENT: Portal SUSPENDED	75	Phishing	100	Phishing	Phishing	✓	C-01 (5 obfusc.)
TC-03	support@insurance-verify.click / Insurance Verification	90	Phishing	90	Phishing	Phishing	✓	—
TC-04	hr@yourclinic.org / Staff meeting notes	10	Safe	10	Safe	Safe	✓	—
TC-05	noreply@123456789@mail.com / Security Alert	80	Phishing	100	Phishing	Phishing	✓	C-01 (2 obfusc.)
TC-06	admin@yourclinic.org / Payroll maintenance	—	—	0	Safe	Safe	✓	New case
TC-07	billing@healthcare-updates.net / Medicare Update	—	—	70	Phishing	Phishing	✓	C-02 spoof
TC-08	noreply@accounts-yourclinic.xyz / Health plan renewal	—	—	40	Suspicious	Phishing	✗	—
TC-09	labs@yourclinic.org / Lab results available	—	—	20	Safe	Suspicious	✗	—

TC-10	no-reply@account-verify.click / V3RIFY YOUR ACC0UNT NOW	—	—	80	Phishing	Phishing	✓	C-01 (3 obfusc.)
TC-11	dr.johnson@yourclinic.org / Referral authorization	—	—	20	Safe	Safe	✓	New case
TC-12	support@login-...secureportal.xyz / Action Required	—	—	100	Phishing	Phishing	✓	C-01+ C-06
TC-13	scheduler@yourclinic.org / Appointment reminder	—	—	0	Safe	Safe	✓	New case
TC-14	claims@insurance-portal.ml / EOB Claim Denied	—	—	100	Phishing	Phishing	✓	C-02+ C-05
TC-15	it@yourclinic.org / Password reminder	—	—	15	Safe	Safe	✓	New case

Gray rows (TC-08, TC-09) = cases where result did not match expected. Both are documented as edge cases rather than system failures — see Section 4.3.

4.2 Accuracy Metrics — v1 Baseline vs. v2 With Controls

Metric	v1 Baseline	v2 With Controls
Test cases run	5	**15**
Passed	5/5	**13/15**
True Positives	3	**8**
True Negatives	2	**6**
False Positives	0	**0**
False Negatives	0	**1**
Precision	100%	**100%**
Recall	est. 60%	**88.9%**
F1 Score	est. 75%	**94.1%**
Accuracy	est. 60%	**93.3%**

Obfuscation detection	No	**Yes**
Display-name detection	No	**Yes**
Quarantine flag	No	**Yes**
Logging	No	**Yes (CSV)**

4.3 Edge Cases & Limitations

TC-08: Subdomain Spoofing — Scored Suspicious, Expected Phishing

The sender noreply@accounts-yourclinic.xyz uses a subdomain that includes "yourclinic" to appear legitimate, but the TLD is .xyz (a known phishing TLD). The system correctly flagged the sender as suspicious (+20) and detected two healthcare lures (+20), producing a score of 40 — which falls in the Suspicious band rather than Phishing. This is a genuine edge case: the system correctly identified the email as a threat, but did not meet the 60-point threshold for quarantine. Residual risk documented in Section 5.

TC-09: Internal Lab Results Email — Scored Safe, Expected Suspicious

The sender labs@yourclinic.org is a trusted internal domain, so no sender flag was triggered. The body mentions "patient portal" and "lab results available" — two healthcare lures that added 20 points, landing at Safe (20/100). The expected classification was Suspicious because the email links to the portal. This reveals a known limitation: the system does not penalize HTTPS portal links from trusted domains, which is correct behavior for a healthcare system. The expected classification in the test case was overly conservative.

5. Updated Risk Register

The following register reflects residual risk after all v2 defensive controls are in place. Baseline scores are carried from Checkpoint 2. Two new risks were identified during v2 validation testing.

Risk	BL L	BL I	BL Scr	Controls Applied	Res L	Res I	Res Scr	Notes
Phishing bypass / AI evasion	5	5	**25**	C-01 Obfuscation detection; expanded keyword library	2	5	**10**	Reduced — obfuscation now caught; sophisticated evasion still possible
Credential theft via fake login	5	5	**25**	C-02 Display-name spoofing; C-06 login URL patterns	2	5	**10**	Reduced — display-name spoofing caught; URL patterns expanded
Malware via attachment	4	5	**20**	Attachment flag (+10 pts); quarantine flag (C-03)	3	5	**15**	Partially reduced — still no deep attachment scanning
AI evasion (adversarial craft)	4	5	**20**	C-01 Obfuscation patterns; broader feature set	3	5	**15**	Partially reduced — obfuscation caught; novel techniques not yet covered
Employee ignores warning	3	3	**9**	Quarantine (C-03) removes decision for Phishing; logs (C-04)	2	3	**6**	Reduced — Phishing emails now quarantined, not just warned

Compromised internal account	2	5	**10**	Display-name trust check partially addresses this	2	5	**10**	Unchanged — internal sender spoofing not fully addressed by current controls
TC-08 subdomain spoof gap	3	4	**12**	Identified in validation — threshold adjustment needed	3	4	**12**	New risk — subdomain spoofing may score Suspicious not Phishing

BL = Baseline (CP2). Res = Residual (post-controls). L = Likelihood (1–5). I = Impact (1–5). Score = L × I.

6. Draft Outline — Final Report

The following structure is planned for the final deliverable. Each section maps directly to work completed across Checkpoints 1, 2, and 3.

Title Page & Executive Summary

- Project overview, scope, key findings, top recommendations
- Written for a non-technical audience (clinic administrator, compliance officer)

Section 1: Introduction & Scope

- Healthcare clinic environment profile and threat context
- Assets in scope: patient data, employee credentials, email system, AI model, internal network
- Assessment methodology: asset identification → threat modeling → risk scoring → control design → validation

Section 2: System Architecture

- Workflow diagram (CP2 Figure 1) with trust boundary annotations
- Description of phishing_detector.py pipeline stages
- Technology decisions and scope constraints

Section 3: Threat Analysis & Risk Baseline (CP1 + CP2)

- Asset inventory with priority ratings
- Threat model: key threats, attack surfaces, abuse scenarios
- CP2 risk matrix with baseline likelihood × impact scores

Section 4: Defensive Controls (CP3)

- C-01 through C-06 implementation details with code
- Control selection rationale mapped to CP2 risks
- Filtering logic, logging design, quarantine design

Section 5: Validation Results (CP3)

- Full 15-case test table with before/after scores
- Accuracy metrics: precision 100%, recall 88.9%, F1 94.1%, accuracy 93.3%
- Edge case analysis (TC-08, TC-09) and documented limitations
- phishing_log.csv as audit trail evidence

Section 6: Updated Risk Register

- Residual risk scores after v2 controls
- Two new risks identified during validation
- Residual risk acceptance statement

Section 7: Recommendations & Roadmap

- Short-term (0–30 days): Push phishing_detector_v2.py to GitHub src/; add threshold tuning for subdomain TLDs
- Medium-term (30–60 days): Integrate logging into persistent database; build minimal warning UI
- Long-term (60+ days): Replace rule-based engine with trained ML classifier; integrate with live email environment simulation

Section 8: Conclusion

- Summary of risk posture improvement from CP2 to CP3
- Key wins: obfuscation detection, display-name spoofing, quarantine, logging
- Remaining gaps and next steps

Appendices

- A: Full phishing_detector_v2.py source code
- B: phishing_log.csv — 15-case test run output
- C: System architecture diagram
- D: CP2 baseline test results (5 cases) for comparison

7. Validation Screenshots

	A	B	C	D	E	F	
1	nestamp	sender	subject	risk_score	classification	quarantined	flags
2	/20/2026 3:42	billing@yourclinic.org	Your appointment is confirmed for Tuesday	0	Safe	FALSE	
3	/20/2026 3:42	no-reply@medicarÐµ-update.xyz	URGENT: Patient portal access will be SUSPENDED	100	Phishing	TRUE	Phishing keyword: 'confirm your identity' | Healthcare lure: 'patient portal' | Suspicious sender: no-re
4	/20/2026 3:42	support@insurance-verify.click	Insurance Verification Required	90	Phishing	TRUE	Phishing keyword: 'update your billing' | Phishing keyword: 'validate your credentials' | Healthcare lur
5	/20/2026 3:42	hr@yourclinic.org	Staff meeting notes attached	10	Safe	FALSE	Attachment present
6	/20/2026 3:42	noreply@123456789@mail.com	Security Alert: Suspicious activity on your account	100	Phishing	TRUE	Phishing keyword: 'reset your password now' | Phishing keyword: 'security alert' | Phishing keyword: '
7	/20/2026 3:42	admin@yourclinic.org	Payroll system maintenance tonight	0	Safe	FALSE	
8	/20/2026 3:42	billing@healthcare-updates.net	Medicare Update: Action Required	70	Phishing	TRUE	Phishing keyword: 'confirm your email address' | Healthcare lure: 'medicare update' | Suspicious URL
9	/20/2026 3:42	noreply@accounts-yourclinic.xyz	Your health plan renewal is pending	40	Suspicious	FALSE	Healthcare lure: 'patient portal' | Healthcare lure: 'health plan renewal' | Suspicious sender: noreply@
10	/20/2026 3:42	labs@yourclinic.org	Lab results available for Patient #4821	20	Safe	FALSE	Healthcare lure: 'patient portal' | Healthcare lure: 'lab results available'
11	/20/2026 3:42	no-reply@account-verify.click	V3RIFY YOUR ACCOUNT NOW	80	Phishing	TRUE	Phishing keyword: 'login attempt' | Suspicious sender: no-reply@account-verify.click | Obfuscation c
12	/20/2026 3:42	dr.johnson@yourclinic.org	Referral authorization for patient follow-up	20	Safe	FALSE	Healthcare lure: 'referral authorization' | Attachment present
13	/20/2026 3:42	support@login-yourclinic.secureportal.xyz	Action Required: Confirm your login within 24 hours	100	Phishing	TRUE	Phishing keyword: 'your account has been compromised' | Phishing keyword: 'action required within
14	/20/2026 3:42	scheduler@yourclinic.org	Reminder: Your appointment tomorrow at 3pm	0	Safe	FALSE	
15	/20/2026 3:42	claims@insurance-portal.ml	EOB Statement: Claim Denied â€" Immediate Response Required	100	Phishing	TRUE	Phishing keyword: 'validate your credentials' | Phishing keyword: 'immediate response required' | Hea

```
PS C:\Users\Owner\Downloads\submission> c:; cd 'c:\Users\Owner\Downloads\submission'; & 'c:\Users\Owner\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\Owner\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '61917' '--' 'C:\Users\Owner\Downloads\submission\phishing_detector_v2.py'

  SECTION 6 — Validation Summary & Before/After Comparison
  Accuracy metrics across all 15 test cases

  ============================================================
  VALIDATION SUMMARY — Checkpoint 3
  ============================================================
  Total test cases   : 15
  Passed             : 13 / 15
  True Positives     : 8  (correctly flagged as threat)
  True Negatives     : 6  (correctly passed as safe)
  False Positives    : 0  (safe email incorrectly flagged)
  False Negatives    : 1  (threat email missed)
  Precision          : 100.00%
  Recall             : 88.89%
  F1 Score           : 94.12%
  Accuracy           : 93.33%

  Log file written   : phishing_log.csv
  ============================================================

  BEFORE vs. AFTER COMPARISON

  Metric              v1 Baseline    v2 With Controls
  ------------------------------------------------------
  Test cases          5              15
  Obfuscation catch   No             Yes (TC-10 caught)
  Display-name spoof  No             Yes (TC-07, TC-14)
  Logging             No             Yes (CSV log)
  Quarantine flag     No             Yes (auto-set)
  URL patterns        4              7
  Keyword list        15             21
  Healthcare lures    8              15
  Accuracy            60.0%          93.3%
  ------------------------------------------------------

PS C:\Users\Owner\Downloads\submission>
```

```
PS C:\Users\Owner\Downloads\submission> c:; cd 'c:\Users\Owner\Downloads\submission'; & 'c:\Users\Owner\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\Owner\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '61917' '--' 'C:\Users\Owner\Downloads\submission\phishing_detector_v2.py'

  SECTION 5 — Edge Cases (TC-08, TC-09)
  Documented limitations — see CP3 report Section 4.3


  ============================================================
  [TC-08] FROM    : noreply@accounts-yourclinic.xyz
  SUBJECT         : Your health plan renewal is pending
  ============================================================
  CLASSIFICATION : Suspicious
  RISK SCORE     : 40/100
  QUARANTINED    : No
  LOGGED         : YES — phishing_log.csv
  ACTION         : Warn user before opening. Flag for IT review.
  FLAGS (3):
    - Healthcare lure: 'patient portal'
    - Healthcare lure: 'health plan renewal'
    - Suspicious sender: noreply@accounts-yourclinic.xyz

  [FAIL] Expected: Phishing | Got: Suspicious — Subdomain spoofing + health plan lure


  ============================================================
  [TC-09] FROM    : lab@yourclinic.org
  SUBJECT         : Lab results available for Patient #4821
  ============================================================
  CLASSIFICATION : Safe
  RISK SCORE     : 20/100
  QUARANTINED    : No
  LOGGED         : YES — phishing_log.csv
  ACTION         : Deliver to inbox.
  FLAGS (2):
    - Healthcare lure: 'patient portal'
    - Healthcare lure: 'lab results available'

  [FAIL] Expected: Suspicious | Got: Safe — Legitimate-looking lab result email — borderline due to portal link
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
PS C:\Users\Owner\Downloads\submission> c:; cd 'c:\Users\Owner\Downloads\submission'; & 'c:\Users\Owner\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\Owner\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '61917' '--' 'C:\Users\Owner\Downloads\submission\phishing_detector_v2.py'

  SECTION 4 — Obfuscation / Evasion Detection (TC-10, TC-12)
  NEW v2 control: C-01 — catches character-substitution evasion


============================================================
 [TC-10] FROM    : no-reply@account-verify.click
 SUBJECT         : V3RIFY YOUR ACCOUNT NOW
============================================================
 CLASSIFICATION : Phishing
 RISK SCORE     : 80/100
 QUARANTINED    : YES
 LOGGED         : YES — phishing_log.csv
 ACTION         : Quarantine immediately. Do not deliver.
 FLAGS (5):
   - Phishing keyword: 'login attempt'
   - Suspicious sender: no-reply@account-verify.click
   - Obfuscation detected: obfuscated 'verify'
   - Obfuscation detected: obfuscated 'account'
   - Obfuscation detected: obfuscated 'click here'

 [PASS] Expected: Phishing | Got: Phishing — Obfuscated keyword evasion attempt — NEW v2 control catches this


============================================================
 [TC-12] FROM    : support@login-yourclinic.secureportal.xyz
 SUBJECT         : Action Required: Confirm your login within 24 hours
============================================================
 CLASSIFICATION : Phishing
 RISK SCORE     : 100/100
 QUARANTINED    : YES
 LOGGED         : YES — phishing_log.csv
 ACTION         : Quarantine immediately. Do not deliver.
 FLAGS (7):
   - Phishing keyword: 'your account has been compromised'
   - Phishing keyword: 'action required within 24 hours'
   - Phishing keyword: 'failure to respond will result'
   - Suspicious sender: support@login-yourclinic.secureportal.xyz
   - Suspicious URL pattern in body
   - Obfuscation detected: obfuscated 'verify'
   - Obfuscation detected: obfuscated 'account'

 [PASS] Expected: Phishing | Got: Phishing — Multi-signal attack: keywords + obfuscation + spoofed URL
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
PS C:\Users\Owner\Downloads\submission> c:; cd 'c:\Users\Owner\Downloads\submission'; & 'c:\Users\Owner\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\Owner\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '61917' '--' 'C:\Users\Owner\Downloads\submission\phishing_detector_v2.py'

  SECTION 3 — Display-Name Spoofing Detection (TC-07, TC-14)
  NEW v2 control: C-02 — catches sender impersonation


============================================================
 [TC-07] FROM    : billing@healthcare-updates.net
 DISPLAY NAME: Clinic Billing Department
 SUBJECT         : Medicare Update: Action Required
============================================================
 CLASSIFICATION : Phishing
 RISK SCORE     : 70/100
 QUARANTINED    : YES
 LOGGED         : YES — phishing_log.csv
 ACTION         : Quarantine immediately. Do not deliver.
 FLAGS (4):
   - Phishing keyword: 'confirm your email address'
   - Healthcare lure: 'medicare update'
   - Suspicious URL pattern in body
   - Display-name spoofing: 'Clinic Billing Department' claims clinic but sender is billing@healthcare-updates.net

 [PASS] Expected: Phishing | Got: Phishing — Display-name spoofing + medicare lure + fake URL


============================================================
 [TC-14] FROM    : claims@insurance-portal.ml
 DISPLAY NAME: HealthFirst Insurance Claims
 SUBJECT         : EOB Statement: Claim Denied — Immediate Response Required
============================================================
 CLASSIFICATION : Phishing
 RISK SCORE     : 100/100
 QUARANTINED    : YES
 LOGGED         : YES — phishing_log.csv
 ACTION         : Quarantine immediately. Do not deliver.
 FLAGS (9):
   - Phishing keyword: 'validate your credentials'
   - Phishing keyword: 'immediate response required'
   - Healthcare lure: 'claim denied'
   - Healthcare lure: 'eob statement'
   - Suspicious sender: claims@insurance-portal.ml
   - Suspicious URL pattern in body
   - Attachment present
   - Display-name spoofing: 'HealthFirst Insurance Claims' claims clinic but sender is claims@insurance-portal.ml

 [PASS] Expected: Phishing | Got: Phishing — High-complexity attack: EOB lure + display-name spoof + all signals
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
PS C:\Users\Owner\Downloads\submission> c:; cd 'c:\Users\Owner\Downloads\submission'; & 'c:\Users\Owner\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\Owner\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '61917' '--' 'C:\Users\Owner\Downloads\submission\phishing_detector_v2.py'

  SECTION 2 — New Legitimate Email Tests (TC-06, TC-11, TC-13, TC-15)
  False-positive checks — safe emails must not be flagged


============================================================
 [TC-06] FROM    : admin@yourclinic.org
 SUBJECT         : Payroll system maintenance tonight
============================================================
 CLASSIFICATION : Safe
 RISK SCORE     : 0/100
 QUARANTINED    : No
 LOGGED         : YES — phishing_log.csv
 ACTION         : Deliver to inbox.

 [PASS] Expected: Safe | Got: Safe — Legitimate internal admin notification


============================================================
 [TC-11] FROM    : dr.johnson@yourclinic.org
 SUBJECT         : Referral authorization for patient follow-up
============================================================
 CLASSIFICATION : Safe
 RISK SCORE     : 20/100
 QUARANTINED    : No
 LOGGED         : YES — phishing_log.csv
 ACTION         : Deliver to inbox.
 FLAGS (2):
   - Healthcare lure: 'referral authorization'
   - Attachment present

 [PASS] Expected: Safe | Got: Safe — Legitimate physician internal referral email


============================================================
 [TC-13] FROM    : scheduler@yourclinic.org
 SUBJECT         : Reminder: Your appointment tomorrow at 3pm
============================================================
 CLASSIFICATION : Safe
 RISK SCORE     : 0/100
 QUARANTINED    : No
 LOGGED         : YES — phishing_log.csv
 ACTION         : Deliver to inbox.

 [PASS] Expected: Safe | Got: Safe — Legitimate appointment reminder — false positive check


============================================================
 [TC-15] FROM    : it@yourclinic.org
 SUBJECT         : Password policy reminder — expires in 30 days
============================================================
 CLASSIFICATION : Safe
 RISK SCORE     : 15/100
 QUARANTINED    : No
 LOGGED         : YES — phishing_log.csv
 ACTION         : Deliver to inbox.
 FLAGS (1):
   - Obfuscation detected: obfuscated 'password'

 [PASS] Expected: Safe | Got: Safe — Legitimate IT password reminder from trusted domain
```

```
PS C:\Users\Owner\Downloads\submission>  c:; cd 'c:\Users\Owner\Downloads\submission'; & 'c:\Users\Owner\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\Owner\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '61917' '--' 'C:\Users\Owner\Downloads\submissi

  ============================================================
   [TC-03] FROM    : support@insurance-verify.click
   SUBJECT     : Insurance Verification Required
  ============================================================
   CLASSIFICATION : Phishing
   RISK SCORE     : 90/100
   QUARANTINED    : YES
   LOGGED         : YES — phishing_log.csv
   ACTION         : Quarantine immediately. Do not deliver.
   FLAGS (6):
     - Phishing keyword: 'update your billing'
     - Phishing keyword: 'validate your credentials'
     - Healthcare lure: 'insurance verification'
     - Suspicious sender: support@insurance-verify.click
     - Suspicious URL pattern in body
     - Attachment present

   [PASS] Expected: Phishing | Got: Phishing — Insurance lure with IP-link and attachment


  ============================================================
   [TC-04] FROM    : hr@yourclinic.org
   SUBJECT     : Staff meeting notes attached
  ============================================================
   CLASSIFICATION : Safe
   RISK SCORE     : 10/100
   QUARANTINED    : No
   LOGGED         : YES — phishing_log.csv
   ACTION         : Deliver to inbox.
   FLAGS (1):
     - Attachment present

   [PASS] Expected: Safe | Got: Safe — Legitimate internal HR email with attachment


  ============================================================
   [TC-05] FROM    : noreply@123456789@mail.com
   SUBJECT     : Security Alert: Suspicious activity on your account
  ============================================================
   CLASSIFICATION : Phishing
   RISK SCORE     : 100/100
   QUARANTINED    : YES
   LOGGED         : YES — phishing_log.csv
   ACTION         : Quarantine immediately. Do not deliver.
   FLAGS (7):
     - Phishing keyword: 'reset your password now'
     - Phishing keyword: 'security alert'
     - Phishing keyword: 'suspicious activity'
     - Phishing keyword: 'immediate response required'
     - Suspicious sender: noreply@123456789@mail.com
     - Obfuscation detected: obfuscated 'account'
     - Obfuscation detected: obfuscated 'password'

   [PASS] Expected: Phishing | Got: Phishing — Malformed sender with credential-theft keywords
```

```
PS C:\Users\Owner\Downloads\submission> ^C
PS C:\Users\Owner\Downloads\submission>  c:; cd 'c:\Users\Owner\Downloads\submission'; & 'c:\Users\Owner\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\Owner\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '61917' '--' 'C:\Users\Owner\Downloads\submission\phishing_detector_v2.py'

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    AI-Assisted Phishing Detection System — CS361  v2.0
    Checkpoint 3: Defensive Controls & Validation
    Healthcare Clinic Email Environment

   NEW IN v2: Logging | Quarantine | Obfuscation Detection
              Display-Name Spoofing | 15 Test Cases

  ┌─────────────────────────────────────────────┐
  │ SECTION 1 — Regression Tests (TC-01 to TC-05)
  │ Carried over from v1 baseline — verifying no regressions
  └─────────────────────────────────────────────┘


  ============================================================
   [TC-01] FROM    : billing@yourclinic.org
   SUBJECT     : Your appointment is confirmed for Tuesday
  ============================================================
   CLASSIFICATION : Safe
   RISK SCORE     : 0/100
   QUARANTINED    : No
   LOGGED         : YES — phishing_log.csv
   ACTION         : Deliver to inbox.

   [PASS] Expected: Safe | Got: Safe — Legitimate clinic appointment email


  ============================================================
   [TC-02] FROM    : no-reply@medicare-update.xyz
   SUBJECT     : URGENT: Patient portal access will be SUSPENDED
  ============================================================
   CLASSIFICATION : Phishing
   RISK SCORE     : 100/100
   QUARANTINED    : YES
   LOGGED         : YES — phishing_log.csv
   ACTION         : Quarantine immediately. Do not deliver.
   FLAGS (10):
     - Phishing keyword: 'confirm your identity'
     - Healthcare lure: 'patient portal'
     - Suspicious sender: no-reply@medicare-update.xyz
     - Suspicious URL pattern in body
     - Urgency signal: 2 all-caps words in subject
     - Obfuscation detected: obfuscated 'verify'
     - Obfuscation detected: obfuscated 'account'
     - Obfuscation detected: obfuscated 'click here'
     - Obfuscation detected: obfuscated 'suspended'
     - Obfuscation detected: obfuscated 'urgent'

   [PASS] Expected: Phishing | Got: Phishing — Classic healthcare portal phishing


  ============================================================
   [TC-03] FROM    : support@insurance-verify.click
   SUBJECT     : Insurance Verification Required
  ============================================================
   CLASSIFICATION : Phishing
   RISK SCORE     : 90/100
   QUARANTINED    : YES
   LOGGED         : YES — phishing_log.csv
   ACTION         : Quarantine immediately. Do not deliver.
   FLAGS (6):
     - Phishing keyword: 'update your billing'
```

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